

Weekly Research Report — Week 5

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Summary of Work Done

Draft revision against Dr. Fiaidhi's comments, and preparation of the codebase and repository for review. This was primarily a writing and consolidation week rather than an experimental one.

Draft revision

Dr. Fiaidhi's comments on the initial draft were worked through. The repository was organised so the notebook is reachable from the GitHub link, with experiment files in a dedicated directory.

Repository: <https://github.com/hkapur1-tech/Graph-Rag-for-clinical-case>

Supervisor feedback received

Feedback on the revised draft was direct and identified three problems:

1. **The draft is insufficiently focused and the writing needs work.** Structure and clarity both flagged.
2. **The analysis does not reach scholarly scale.** Important hypotheses are stated but tested on 20 random cases.
3. **Reviewer comments should not be deleted** when revising, so that the revision can be checked against them.

The second point is the substantive one and confirms the limitation identified in Week 4. The gap between the strength of the hypotheses and the size of the evidence is the central weakness of the draft as it stands.

Assessment

The 20-question pilot established that the pipeline works end to end, but it cannot support any claim about the relative merit of the architectures. A difference of one question is 5 percentage points. Every accuracy reported so far is within noise of every other.

Scaling to the complete 1,273-question MedQA test set is therefore not an incremental improvement but a precondition for the paper having a result at all. This is the priority for Week 6, along with paired significance testing so that differences can be reported with a stated confidence rather than asserted.

Plan for Week 6

1. Scale evaluation to the full 1,273-question MedQA-USMLE test set, all five architectures
2. Implement McNemar's paired significance test and 95% Wilson confidence intervals
3. Test whether knowledge-graph scale affects the outcome — build a substantially larger graph and compare
4. Rewrite Abstract, Introduction, Methodology, Results and Conclusion against whatever the full-scale numbers show

Blockers

Full-scale evaluation of five architectures over 1,273 questions is roughly 6,000 model calls. Checkpointing will be implemented so that an interrupted run resumes rather than restarting.